Self Check

ⓐ *After completing the exercises, use this checklist to evaluate your mastery of the objectives of this section.*

I can...	Confidently	With some help	No-I don't get it!
solve coin word problems.			
solve ticket and stamp word problems.			

ⓑ *After reviewing this checklist, what will you do to become confident for all objectives?*

9.3 Use Properties of Angles, Triangles, and the Pythagorean Theorem

Learning Objectives

By the end of this section, you will be able to:

- Use the properties of angles
- Use the properties of triangles
- Use the Pythagorean Theorem

☑ **BE PREPARED** 9.7 Before you get started, take this readiness quiz.

Solve: $x + 3 + 6 = 11$.
If you missed this problem, review Example 8.6.

☑ **BE PREPARED** 9.8 Solve: $\frac{a}{45} = \frac{4}{3}$.
If you missed this problem, review Example 6.42.

☑ **BE PREPARED** 9.9 Simplify: $\sqrt{36 + 64}$.
If you missed this problem, review Example 5.72.

So far in this chapter, we have focused on solving word problems, which are similar to many real-world applications of algebra. In the next few sections, we will apply our problem-solving strategies to some common geometry problems.

Use the Properties of Angles

Are you familiar with the phrase 'do a 180'? It means to turn so that you face the opposite direction. It comes from the fact that the measure of an angle that makes a straight line is 180 degrees. See Figure 9.5.

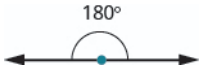


Figure 9.5

An **angle** is formed by two rays that share a common endpoint. Each ray is called a side of the angle and the common endpoint is called the **vertex**. An angle is named by its vertex. In Figure 9.6, $\angle A$ is the angle with vertex at point A. The measure of $\angle A$ is written $m\angle A$.

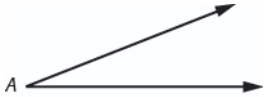


Figure 9.6 $\angle A$ is the angle with vertex at point A.

We measure angles in degrees, and use the symbol $^\circ$ to represent degrees. We use the abbreviation m for the *measure* of an angle. So if $\angle A$ is 27°, we would write $m\angle A = 27$.

If the sum of the measures of two angles is 180°, then they are called **supplementary angles**. In Figure 9.7, each pair of angles is supplementary because their measures add to 180°. Each angle is the *supplement* of the other.

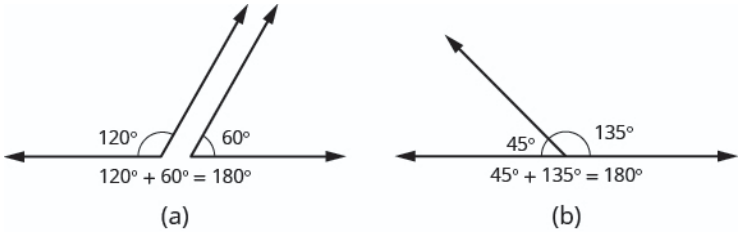


Figure 9.7 The sum of the measures of supplementary angles is $180°$.

If the sum of the measures of two angles is $90°$, then the angles are **complementary angles**. In Figure 9.8, each pair of angles is complementary, because their measures add to $90°$. Each angle is the *complement* of the other.

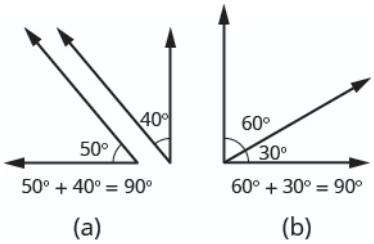


Figure 9.8 The sum of the measures of complementary angles is $90°$.

Supplementary and Complementary Angles

If the sum of the measures of two angles is $180°$, then the angles are supplementary.

If $\angle A$ and $\angle B$ are supplementary, then $m\angle A + m\angle B = 180°$.

If the sum of the measures of two angles is $90°$, then the angles are complementary.

If $\angle A$ and $\angle B$ are complementary, then $m\angle A + m\angle B = 90°$.

In this section and the next, you will be introduced to some common geometry formulas. We will adapt our Problem Solving Strategy for Geometry Applications. The geometry formula will name the variables and give us the equation to solve.

In addition, since these applications will all involve geometric shapes, it will be helpful to draw a figure and then label it with the information from the problem. We will include this step in the Problem Solving Strategy for Geometry Applications.

HOW TO

Use a Problem Solving Strategy for Geometry Applications.

Step 1. **Read** the problem and make sure you understand all the words and ideas. Draw a figure and label it with the given information.
Step 2. **Identify** what you are looking for.
Step 3. **Name** what you are looking for and choose a variable to represent it.
Step 4. **Translate** into an equation by writing the appropriate formula or model for the situation. Substitute in the given information.
Step 5. **Solve** the equation using good algebra techniques.
Step 6. **Check** the answer in the problem and make sure it makes sense.
Step 7. **Answer** the question with a complete sentence.

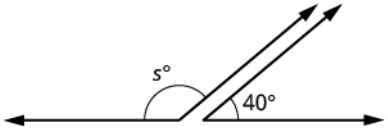
The next example will show how you can use the Problem Solving Strategy for Geometry Applications to answer questions about supplementary and complementary angles.

EXAMPLE 9.16

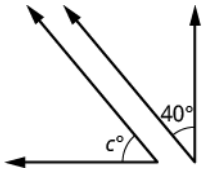
An angle measures $40°$. Find (a) its supplement, and (b) its complement.

Solution

(a)

Step 1. **Read** the problem. Draw the figure and label it with the given information.	
Step 2. **Identify** what you are looking for.	the supplement of a 40° angle.
Step 3. **Name.** Choose a variable to represent it.	let s = the measure of the supplement
Step 4. **Translate.** Write the appropriate formula for the situation and substitute in the given information.	$m\angle A + m\angle B = 180$ $s + 40 = 180$
Step 5. **Solve** the equation.	$s = 140$
Step 6. **Check:** $140 + 40 \stackrel{?}{=} 180$ $180 = 180$ ✓	
Step 7. **Answer** the question.	The supplement of the 40° angle is 140°.

(b)

Step 1. **Read** the problem. Draw the figure and label it with the given information.	
Step 2. **Identify** what you are looking for.	the complement of a 40° angle.
Step 3. **Name.** Choose a variable to represent it.	let c = the measure of the complement
Step 4. **Translate.** Write the appropriate formula for the situation and substitute in the given information.	$m\angle A + m\angle B = 90$
Step 5. **Solve** the equation.	$c + 40 = 90$ $c = 50$

Step 6. **Check:**

$$50 + 40 \stackrel{?}{=} 90$$
$$90 = 90 \checkmark$$

Step 7. **Answer** the question.	The complement of the 40° angle is 50°.

TRY IT 9.31 An angle measures $25°$. Find its: (a) supplement (b) complement.

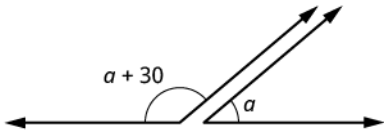
TRY IT 9.32 An angle measures $77°$. Find its: (a) supplement (b) complement.

Did you notice that the words complementary and supplementary are in alphabetical order just like 90 and 180 are in numerical order?

EXAMPLE 9.17

Two angles are supplementary. The larger angle is $30°$ more than the smaller angle. Find the measure of both angles.

Solution

Step 1. **Read** the problem. Draw the figure and label it with the given information.	
Step 2. **Identify** what you are looking for.	the measures of both angles
Step 3. **Name.** Choose a variable to represent it. The larger angle is 30° more than the smaller angle.	let a = measure of smaller angle $a + 30$ = measure of larger angle
Step 4. **Translate.** Write the appropriate formula and substitute.	$m\angle A + m\angle B = 180$
Step 5. **Solve** the equation.	$(a + 30) + a = 180$ $2a + 30 = 180$ $2a = 150$ $a = 75$ measure of smaller angle $a + 30$ measure of larger angle $75 + 30$ 105
Step 6. **Check:** $m\angle A + m\angle B = 180$ $75 + 105 \stackrel{?}{=} 180$ $180 = 180 \checkmark$	
Step 7. **Answer** the question.	The measures of the angles are 75° and 105°.

TRY IT 9.33 Two angles are supplementary. The larger angle is $100°$ more than the smaller angle. Find the measures of both angles.

TRY IT 9.34 Two angles are complementary. The larger angle is $40°$ more than the smaller angle. Find the measures of both angles.

Use the Properties of Triangles

What do you already know about triangles? Triangle have three sides and three angles. Triangles are named by their vertices. The **triangle** in Figure 9.9 is called ΔABC, read 'triangle ABC'. We label each side with a lower case letter to match the upper case letter of the opposite vertex.

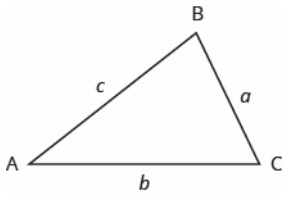


Figure 9.9 ΔABC has vertices A, B, and C and sides a, b, and c.

The three angles of a triangle are related in a special way. The sum of their measures is $180°$.

$$m\angle A + m\angle B + m\angle C = 180°$$

Sum of the Measures of the Angles of a Triangle

For any ΔABC, the sum of the measures of the angles is $180°$.

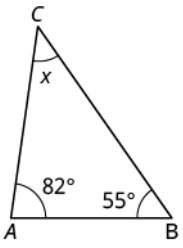
$$m\angle A + m\angle B + m\angle C = 180°$$

EXAMPLE 9.18

The measures of two angles of a triangle are $55°$ and $82°$. Find the measure of the third angle.

Solution

Step 1. **Read** the problem. Draw the figure and label it with the given information.



Step 2. **Identify** what you are looking for.	the measure of the third angle in a triangle
Step 3. **Name.** Choose a variable to represent it.	let x = the measure of the angle
Step 4. **Translate.** Write the appropriate formula and substitute.	$m\angle A + m\angle B + m\angle C = 180$

Step 5. **Solve** the equation.	$55+82+x=180$ $137+x=180$ $x=43$
Step 6. **Check:** $55+82+43\stackrel{?}{=}180$ $180=180$ ✓	
Step 7. **Answer** the question.	The measure of the third angle is 43 degrees.

TRY IT 9.35 The measures of two angles of a triangle are $31°$ and $128°$. Find the measure of the third angle.

TRY IT 9.36 A triangle has angles of $49°$ and $75°$. Find the measure of the third angle.

Right Triangles

Some triangles have special names. We will look first at the **right triangle**. A right triangle has one $90°$ angle, which is often marked with the symbol shown in Figure 9.10.

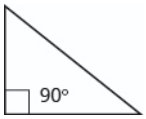


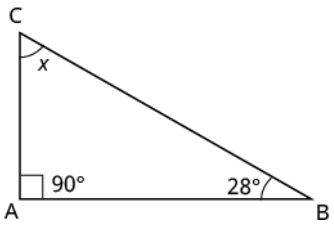
Figure 9.10

If we know that a triangle is a right triangle, we know that one angle measures $90°$ so we only need the measure of one of the other angles in order to determine the measure of the third angle.

EXAMPLE 9.19

One angle of a right triangle measures $28°$. What is the measure of the third angle?

Solution

Step 1. **Read** the problem. Draw the figure and label it with the given information.	
Step 2. **Identify** what you are looking for.	the measure of an angle
Step 3. **Name.** Choose a variable to represent it.	let x = the measure of the angle
Step 4. **Translate.** Write the appropriate formula and substitute.	$m\angle A+m\angle B+m\angle C=180$

Step 5. **Solve** the equation.	$x + 90 + 28 = 180$ $x + 118 = 180$ $x = 62$
Step 6. **Check:** $180 \stackrel{?}{=} 90 + 28 + 62$ $180 = 180$ ✓	
Step 7. **Answer** the question.	The measure of the third angle is 62°.

TRY IT 9.37 One angle of a right triangle measures $56°$. What is the measure of the other angle?

TRY IT 9.38 One angle of a right triangle measures $45°$. What is the measure of the other angle?

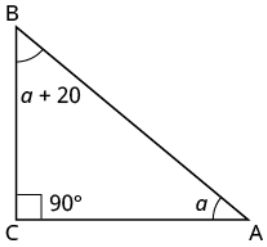
In the examples so far, we could draw a figure and label it directly after reading the problem. In the next example, we will have to define one angle in terms of another. So we will wait to draw the figure until we write expressions for all the angles we are looking for.

EXAMPLE 9.20

The measure of one angle of a right triangle is $20°$ more than the measure of the smallest angle. Find the measures of all three angles.

Solution

Step 1. **Read** the problem.

Step 2. **Identify** what you are looking for.	the measures of all three angles
Step 3. **Name.** Choose a variable to represent it. Now draw the figure and label it with the given information.	Let a = 1st angle $a + 20$ = 2nd angle 90 = 3rd angle (the right angle) 
Step 4. **Translate.** Write the appropriate formula and substitute into the formula.	$m\angle A + m\angle B + m\angle C = 180$ $a + (a + 20) + 90 = 180$

Step 5. **Solve** the equation.	$2a + 110 = 180$ $2a = 70$ $a = 35$ first angle $a + 20$ second angle $35 + 20$ 55 90 third angle
Step 6. **Check:** $35 + 55 + 90 \stackrel{?}{=} 180$ $180 = 180$ ✓	
Step 7. **Answer** the question.	The three angles measure 35°, 55°, and 90°.

TRY IT 9.39 The measure of one angle of a right triangle is $50°$ more than the measure of the smallest angle. Find the measures of all three angles.

TRY IT 9.40 The measure of one angle of a right triangle is $30°$ more than the measure of the smallest angle. Find the measures of all three angles.

Similar Triangles

When we use a map to plan a trip, a sketch to build a bookcase, or a pattern to sew a dress, we are working with similar figures. In geometry, if two figures have exactly the same shape but different sizes, we say they are **similar figures**. One is a scale model of the other. The corresponding sides of the two figures have the same ratio, and all their corresponding angles have the same measures.

The two triangles in Figure 9.11 are similar. Each side of ΔABC is four times the length of the corresponding side of ΔXYZ and their corresponding angles have equal measures.

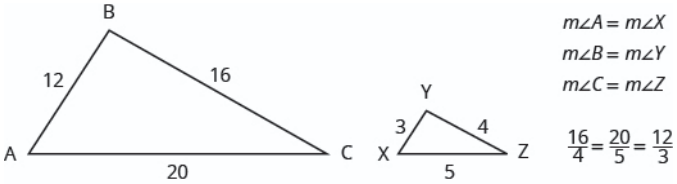
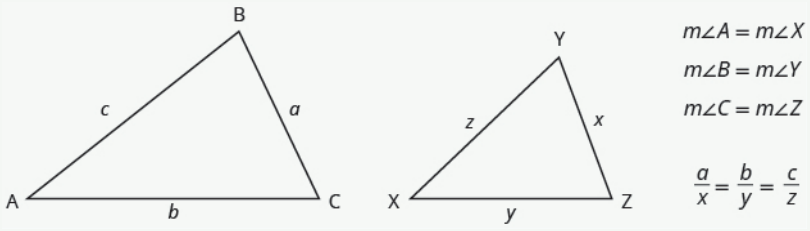


Figure 9.11 ΔABC and ΔXYZ are similar triangles. Their corresponding sides have the same ratio and the corresponding angles have the same measure.

Properties of Similar Triangles

If two triangles are similar, then their corresponding angle measures are equal and their corresponding side lengths are in the same ratio.



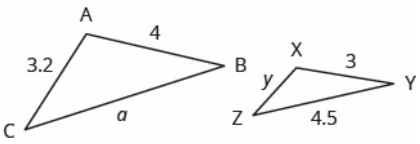
The length of a side of a triangle may be referred to by its endpoints, two vertices of the triangle. For example, in ΔABC:

the length a can also be written BC
the length b can also be written AC
the length c can also be written AB

We will often use this notation when we solve similar triangles because it will help us match up the corresponding side lengths.

EXAMPLE 9.21

ΔABC and ΔXYZ are similar triangles. The lengths of two sides of each triangle are shown. Find the lengths of the third side of each triangle.



Solution

Step 1. **Read** the problem. Draw the figure and label it with the given information.	The figure is provided.
Step 2. **Identify** what you are looking for.	The length of the sides of similar triangles
Step 3. **Name.** Choose a variable to represent it.	Let a = length of the third side of ΔABC y = length of the third side ΔXYZ
Step 4. **Translate.**	

The triangles are similar, so the corresponding sides are in the same ratio. So

$$\frac{AB}{XY} = \frac{BC}{YZ} = \frac{AC}{XZ}$$

Since the side $AB = 4$ corresponds to the side $XY = 3$, we will use the ratio $\frac{AB}{XY} = \frac{4}{3}$ to find the other sides.

Be careful to match up corresponding sides correctly.

	To find a:	To find y:
sides of large triangle →	$\frac{AB}{XY} = \frac{BC}{YZ}$	$\frac{AB}{XY} = \frac{AC}{XZ}$
sides of small triangle →	$\frac{4}{3} = \frac{a}{4.5}$	$\frac{4}{3} = \frac{3.2}{y}$

Step 5. **Solve** the equation.	$3a = 4(4.5)$ $3a = 18$ $a = 6$	$4y = 3(3.2)$ $4y = 9.6$ $y = 2.4$

Step 6. **Check:**

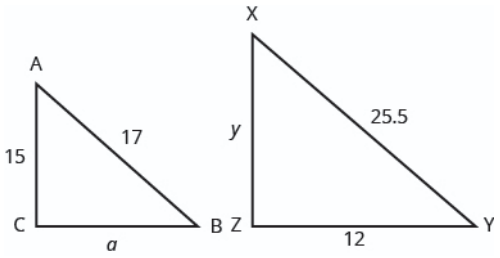
$$\frac{4}{3} \stackrel{?}{=} \frac{6}{4.5} \qquad \frac{4}{3} \stackrel{?}{=} \frac{3.2}{2.4}$$

$$4(4.5) \stackrel{?}{=} 6(3) \qquad 4(2.4) \stackrel{?}{=} 3.2(3)$$

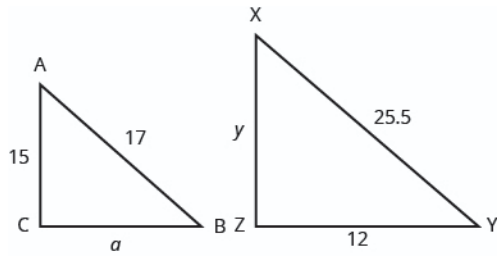
$$18 = 18 \checkmark \qquad 9.6 = 9.6 \checkmark$$

Step 7. **Answer** the question.	The third side of ΔABC is 6 and the third side of ΔXYZ is 2.4.

TRY IT 9.41 ΔABC is similar to ΔXYZ. Find a.



TRY IT 9.42 ΔABC is similar to ΔXYZ. Find y.



Use the Pythagorean Theorem

The **Pythagorean Theorem** is a special property of right triangles that has been used since ancient times. It is named after the Greek philosopher and mathematician Pythagoras who lived around 500 BCE.

Remember that a right triangle has a $90°$ angle, which we usually mark with a small square in the corner. The side of the triangle opposite the $90°$ angle is called the **hypotenuse**, and the other two sides are called the **legs**. See Figure 9.12.

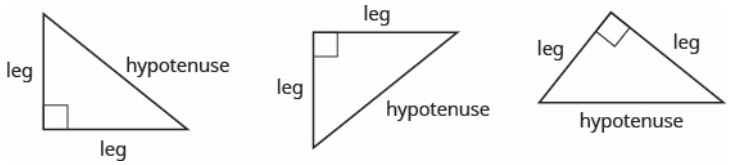


Figure 9.12 In a right triangle, the side opposite the $90°$ angle is called the hypotenuse and each of the other sides is called a leg.

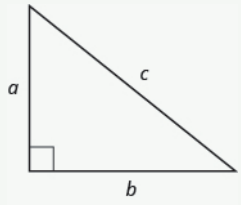
The Pythagorean Theorem tells how the lengths of the three sides of a right triangle relate to each other. It states that in any right triangle, the sum of the squares of the two legs equals the square of the hypotenuse.

The Pythagorean Theorem

In any right triangle ΔABC,

$$a^2 + b^2 = c^2$$

where c is the length of the hypotenuse a and b are the lengths of the legs.



To solve problems that use the Pythagorean Theorem, we will need to find square roots. In Simplify and Use Square Roots we introduced the notation $\sqrt{m}$ and defined it in this way:

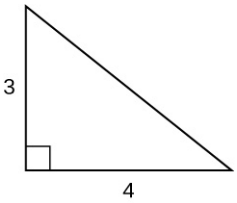
$$\text{If } m = n^2, \text{ then } \sqrt{m} = n \text{ for } n \geq 0$$

For example, we found that $\sqrt{25}$ is 5 because $5^2 = 25$.

We will use this definition of square roots to solve for the length of a side in a right triangle.

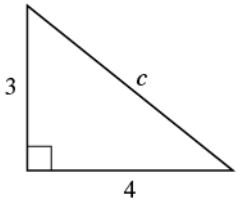
EXAMPLE 9.22

Use the Pythagorean Theorem to find the length of the hypotenuse.



Solution

Step 1. **Read** the problem.

Step 2. **Identify** what you are looking for.	the length of the hypotenuse of the triangle
Step 3. **Name.** Choose a variable to represent it.	Let c = the length of the hypotenuse 
Step 4. **Translate.** Write the appropriate formula. Substitute.	$a^2 + b^2 = c^2$ $3^2 + 4^2 = c^2$
Step 5. **Solve** the equation.	$9 + 16 = c^2$ $25 = c^2$ $\sqrt{25} = c$ $5 = c$

Step 6. **Check:**

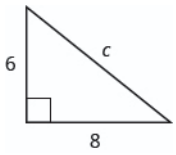
$3^2 + 4^2 = 5^2$

$9 + 16 \stackrel{?}{=} 25$

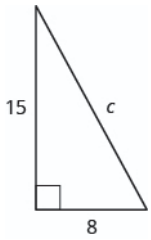
$25 = 25$ ✓

Step 7. **Answer** the question.	The length of the hypotenuse is 5.

TRY IT 9.43 Use the Pythagorean Theorem to find the length of the hypotenuse.

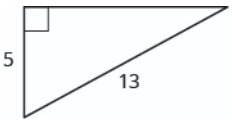


TRY IT 9.44 Use the Pythagorean Theorem to find the length of the hypotenuse.



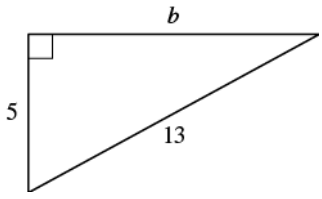
EXAMPLE 9.23

Use the Pythagorean Theorem to find the length of the longer leg.



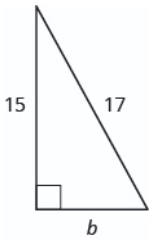
Solution

Step 1. **Read** the problem.

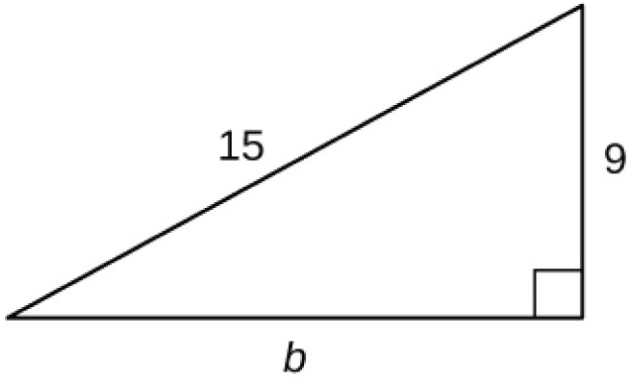
Step 2. **Identify** what you are looking for.	The length of the leg of the triangle
Step 3. **Name.** Choose a variable to represent it.	Let b = the leg of the triangle Label side b 

Step 4. **Translate.** Write the appropriate formula. Substitute.	$a^2 + b^2 = c^2$ $5^2 + b^2 = 13^2$
Step 5. **Solve** the equation. Isolate the variable term. Use the definition of the square root. Simplify.	$25 + b^2 = 169$ $b^2 = 144$ $b = \sqrt{144}$ $b = 12$
Step 6. **Check:** $5^2 + 12^2 \stackrel{?}{=} 13^2$ $25 + 144 \stackrel{?}{=} 169$ $169 = 169$ ✓	
Step 7. **Answer** the question.	The length of the leg is 12.

TRY IT 9.45 Use the Pythagorean Theorem to find the length of the leg.

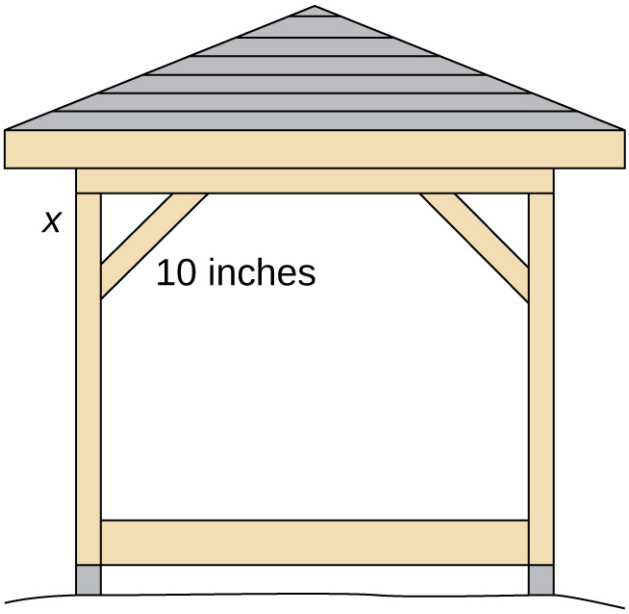


TRY IT 9.46 Use the Pythagorean Theorem to find the length of the leg.



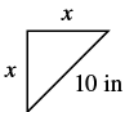
EXAMPLE 9.24

Kelvin is building a gazebo and wants to brace each corner by placing a 10-inch wooden bracket diagonally as shown. How far below the corner should he fasten the bracket if he wants the distances from the corner to each end of the bracket to be equal? Approximate to the nearest tenth of an inch.



Solution

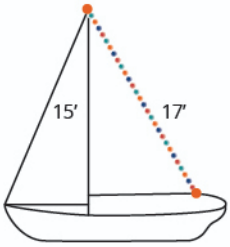
Step 1. **Read** the problem.

Step 2. **Identify** what you are looking for.	the distance from the corner that the bracket should be attached
Step 3. **Name.** Choose a variable to represent it.	Let x = the distance from the corner 
Step 4. **Translate.** Write the appropriate formula. Substitute.	$a^2 + b^2 = c^2$ $x^2 + x^2 = 10^2$
Step 5. **Solve** the equation. Isolate the variable. Use the definition of the square root. Simplify. Approximate to the nearest tenth.	$2x^2 = 100$ $x^2 = 50$ $\sqrt{x^2} = x = \sqrt{50}$ $b \approx 7.1$
Step 6. **Check:** $a^2 + b^2 = c^2$ $(7.1)^2 + (7.1)^2 \overset{?}{\approx} 10^2$ Yes.	
Step 7. **Answer** the question.	Kelvin should fasten each piece of wood approximately 7.1" from the corner.

TRY IT 9.47 John puts the base of a 13-ft ladder 5 feet from the wall of his house. How far up the wall does the ladder reach?



TRY IT 9.48 Randy wants to attach a 17-ft string of lights to the top of the 15-ft mast of his sailboat. How far from the base of the mast should he attach the end of the light string?



MEDIA

ACCESS ADDITIONAL ONLINE RESOURCES

Animation: The Sum of the Interior Angles of a Triangle (https://www.openstax.org/l/24sumintangles)

Similar Polygons (https://www.openstax.org/l/24simpolygons)

Example: Determine the Length of the Hypotenuse of a Right Triangle (https://www.openstax.org/l/24hyporighttri)

SECTION 9.3 EXERCISES

Practice Makes Perfect

Use the Properties of Angles

In the following exercises, find ⓐ the supplement and ⓑ the complement of the given angle.

81. $53°$

82. $16°$

83. $29°$

84. $72°$

In the following exercises, use the properties of angles to solve.

85. Find the supplement of a $135°$ angle.

86. Find the complement of a $38°$ angle.

87. Find the complement of a $27.5°$ angle.

88. Find the supplement of a $109.5°$ angle.

89. Two angles are supplementary. The larger angle is $56°$ more than the smaller angle. Find the measures of both angles.

90. Two angles are supplementary. The smaller angle is $36°$ less than the larger angle. Find the measures of both angles.

91. Two angles are complementary. The smaller angle is $34°$ less than the larger angle. Find the measures of both angles.

92. Two angles are complementary. The larger angle is $52°$ more than the smaller angle. Find the measures of both angles.

Use the Properties of Triangles

In the following exercises, solve using properties of triangles.

93. The measures of two angles of a triangle are $26°$ and $98°$. Find the measure of the third angle.

94. The measures of two angles of a triangle are $61°$ and $84°$. Find the measure of the third angle.

95. The measures of two angles of a triangle are $105°$ and $31°$. Find the measure of the third angle.

96. The measures of two angles of a triangle are $47°$ and $72°$. Find the measure of the third angle.

97. One angle of a right triangle measures $33°$. What is the measure of the other angle?

98. One angle of a right triangle measures $51°$. What is the measure of the other angle?

99. One angle of a right triangle measures $22.5°$. What is the measure of the other angle?

100. One angle of a right triangle measures $36.5°$. What is the measure of the other angle?

101. The two smaller angles of a right triangle have equal measures. Find the measures of all three angles.

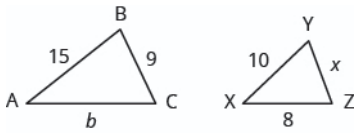
102. The measure of the smallest angle of a right triangle is $20°$ less than the measure of the other small angle. Find the measures of all three angles.

103. The angles in a triangle are such that the measure of one angle is twice the measure of the smallest angle, while the measure of the third angle is three times the measure of the smallest angle. Find the measures of all three angles.

104. The angles in a triangle are such that the measure of one angle is $20°$ more than the measure of the smallest angle, while the measure of the third angle is three times the measure of the smallest angle. Find the measures of all three angles.

Find the Length of the Missing Side

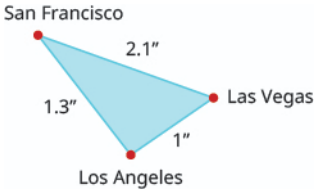
In the following exercises, ΔABC is similar to ΔXYZ. Find the length of the indicated side.



105. side b

106. side x

On a map, San Francisco, Las Vegas, and Los Angeles form a triangle whose sides are shown in the figure below. The actual distance from Los Angeles to Las Vegas is 270 miles.



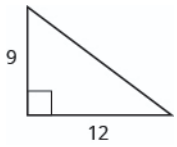
107. Find the distance from Los Angeles to San Francisco.

108. Find the distance from San Francisco to Las Vegas.

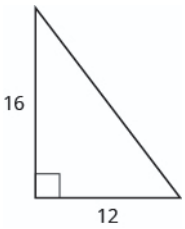
Use the Pythagorean Theorem

In the following exercises, use the Pythagorean Theorem to find the length of the hypotenuse.

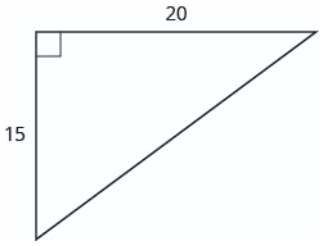
109.



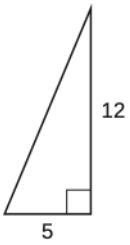
110.



111.



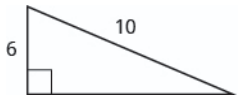
112.



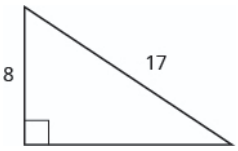
Find the Length of the Missing Side

In the following exercises, use the Pythagorean Theorem to find the length of the missing side. Round to the nearest tenth, if necessary.

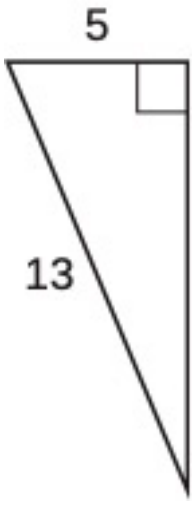
113.



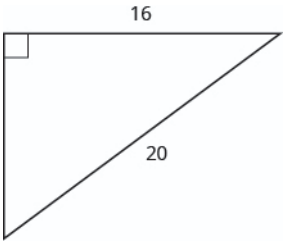
114.



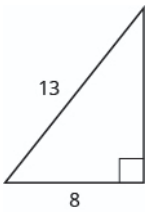
115.



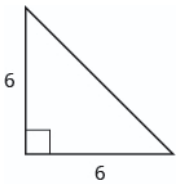
116.



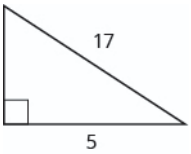
117.



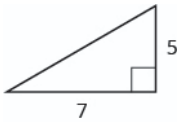
118.



119.

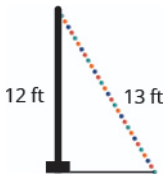


120.



In the following exercises, solve. Approximate to the nearest tenth, if necessary.

121. A 13-foot string of lights will be attached to the top of a 12-foot pole for a holiday display. How far from the base of the pole should the end of the string of lights be anchored?



122. Pam wants to put a banner across her garage door to congratulate her son on his college graduation. The garage door is 12 feet high and 16 feet wide. How long should the banner be to fit the garage door?



123. Chi is planning to put a path of paving stones through her flower garden. The flower garden is a square with sides of 10 feet. What will the length of the path be?



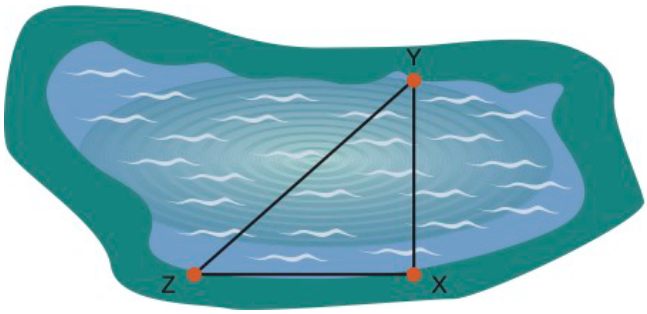
124. Brian borrowed a 20-foot extension ladder to paint his house. If he sets the base of the ladder 6 feet from the house, how far up will the top of the ladder reach?



Everyday Math

125. **Building a scale model** Joe wants to build a doll house for his daughter. He wants the doll house to look just like his house. His house is 30 feet wide and 35 feet tall at the highest point of the roof. If the dollhouse will be 2.5 feet wide, how tall will its highest point be?

126. **Measurement** A city engineer plans to build a footbridge across a lake from point X to point Y, as shown in the picture below. To find the length of the footbridge, she draws a right triangle XYZ, with right angle at X. She measures the distance from X to Z, 800 feet, and from Y to Z, $1{,}000$ feet. How long will the bridge be?



Writing Exercises

127. Write three of the properties of triangles from this section and then explain each in your own words.

128. Explain how the figure below illustrates the Pythagorean Theorem for a triangle with legs of length 3 and 4.

